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EXPERIMENT NO - 2

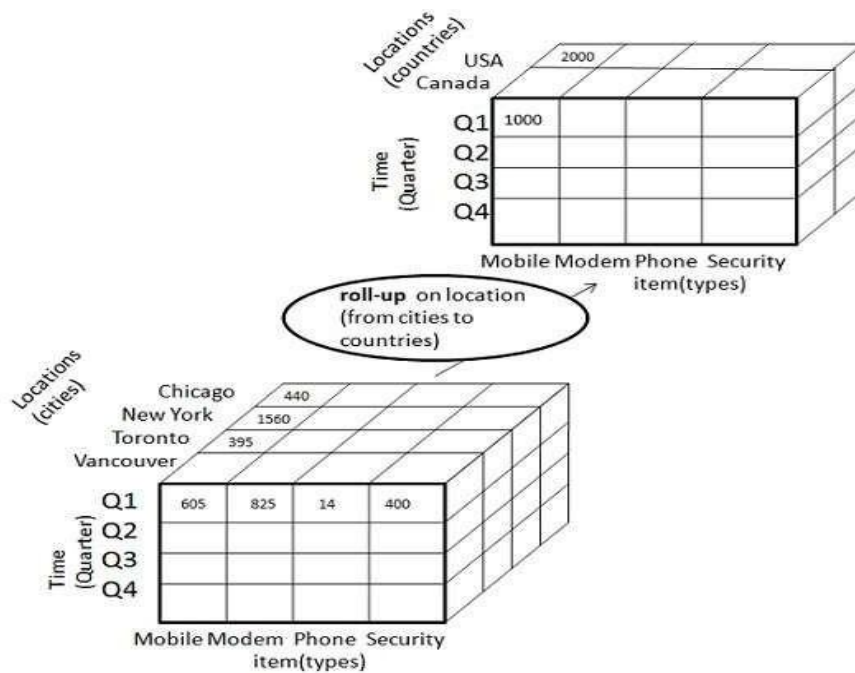
2.1 Aim: Implementation of OLAP operations: Slice, Dice, Rollup, Drilldown and Pivot based on experiment 1.

2.2 Related Theory: OLAP Operations:

Since OLAP servers are based on multidimensional view of data. Here is the list of OLAP operations which can be performed on multidimensional data.

Roll Up

Roll-up performs aggregation on a data cube. The following diagram illustrates how roll-up works.



- Roll-up is performed by climbing up a concept hierarchy for the dimension location.
- Initially the concept hierarchy was "street < city < province < country".
- On rolling up, the data is aggregated by ascending the location hierarchy from the level of city to the level of country.

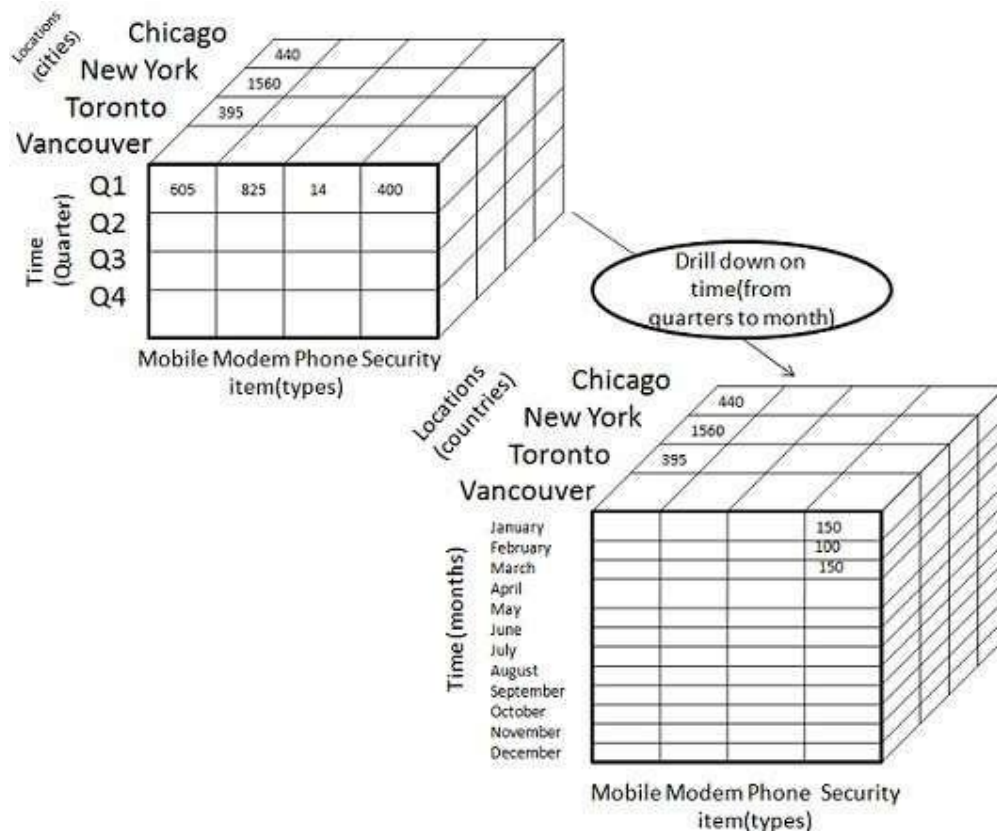
- The data is grouped into cities rather than countries.
- When roll-up is performed, one or more dimensions from the data cube are removed.

Drill Down

Drill-down is the reverse operation of roll-up. It is performed by either of the following ways:

- By stepping down a concept hierarchy for a dimension
- By introducing a new dimension.

The following diagram illustrates how drill-down works:

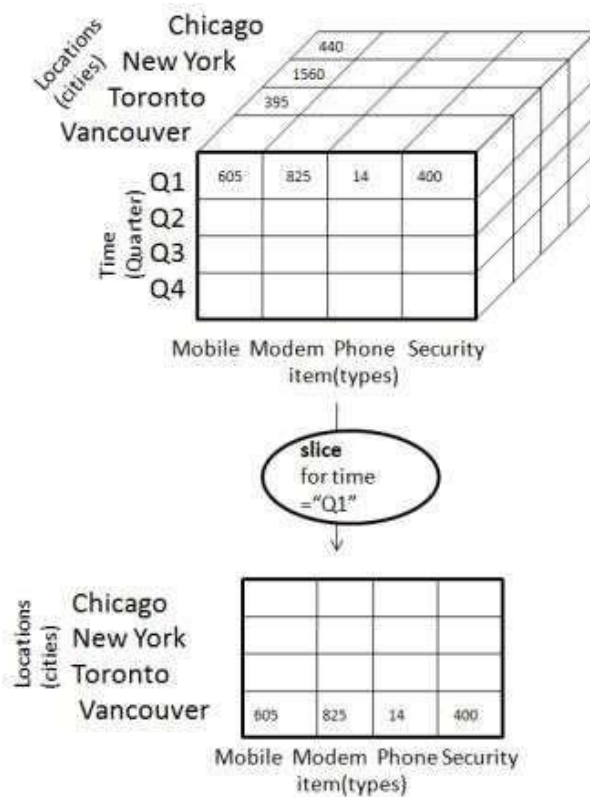


- Drill-down is performed by stepping down a concept hierarchy for the dimension time.
- Initially the concept hierarchy was "day < month < quarter < year."
- On drilling down, the time dimension is descended from the level of quarter to the level of month.
- When drill-down is performed, one or more dimensions from the data cube are added.
- It navigates the data from less detailed data to highly detailed data.

Slice

The slice operation selects one particular dimension from a given cube and provides a new sub-cube.

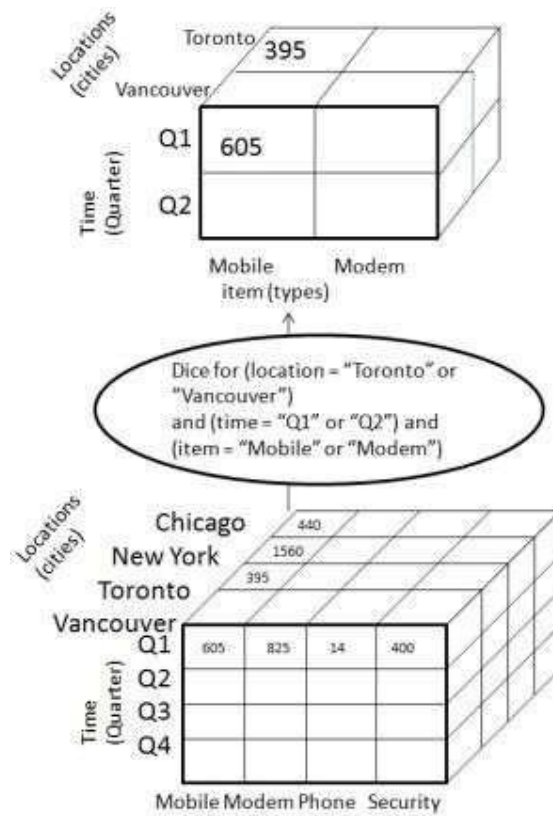
Consider the following diagram that shows how slice works.



- Here Slice is performed for the dimension "time" using the criterion time = "Q1".
- It will form a new sub-cube by selecting one or more dimensions.

Dice

Dice selects two or more dimensions from a given cube and provides a new sub-cube. Consider the following diagram that shows the dice operation.



The dice operation on the cube based on the following selection criteria involves three dimensions.

- (location = "Toronto" or "Vancouver")
- (time = "Q1" or "Q2")
- (item = " Mobile" or "Modem")

Pivot

The pivot operation is also known as rotation. It rotates the data axes in view in order to provide an alternative presentation of data. Consider the following diagram that shows the pivot operation.



In this the item and location axes in 2-D slice are rotated.

2.3 Procedure:

OLAP (Online Analytical Processing) operations allow users to interactively analyze multidimensional data from different perspectives. The following steps outline how to perform each key OLAP operation using the sales data warehouse created in Experiment 1.

1. Roll-Up Operation

Purpose: To aggregate data by moving up the hierarchy in a dimension.

Steps:

- Identify the dimension with a defined hierarchy (e.g., **Location**: Street → City → State → Country).
- Use GROUP BY in SQL to aggregate data at a higher level.

Example:

SELECT

```
l.country,  
SUM(f.Total_sales) AS total_sales  
FROM  
Fact_sales f  
JOIN  
Location_dw l ON f.loc_id = l.loc_id  
GROUP BY  
l.country;
```

2. Drill-Down Operation

Purpose: To get more detailed data by descending a hierarchy or adding a new dimension.

Steps:

- Choose a dimension to drill down (e.g., **Time**: Year → Quarter → Month → Day).
- Modify the query to include a more detailed level.

Example:

```
SELECT  
t.month,  
l.city,  
SUM(f.Total_sales) AS total_sales  
FROM  
Fact_sales f  
JOIN  
Time_dw t ON f.time_id = t.time_id  
JOIN  
Location_dw l ON f.loc_id = l.loc_id  
GROUP BY  
t.month, l.city;
```

3. Slice Operation

Purpose: To extract a sub-cube by fixing one dimension to a specific value.

Steps:

- Select a fixed value for one dimension (e.g., time = 'Q1').
- Use WHERE clause to filter that dimension.

Example:

```
SELECT
    p.Prod_name,
    l.city,
    f.Total_sales
FROM
    Fact_sales f
JOIN
    Product_dw p ON f.prod_id = p.prod_id
JOIN
    Time_dw t ON f.time_id = t.time_id
JOIN
    Location_dw l ON f.loc_id = l.loc_id
WHERE
    t.qt = 'Q1';
```

4. Dice Operation

Purpose: To extract a sub-cube based on multiple dimension filters.

Steps:

- Apply multiple filter conditions across different dimensions.

Example:

```
SELECT
    p.Prod_name,
    t.month,
    l.city,
    f.Total_sales
FROM
    Fact_sales f
JOIN
    Product_dw p ON f.prod_id = p.prod_id
```



```

JOIN
    Time_dw t ON f.time_id = t.time_id
JOIN
    Location_dw l ON f.loc_id = l.loc_id
WHERE
    (l.city = 'MUMBAI' OR l.city = 'KOLKATA') AND
    (t.qt = 'Q1' OR t.qt = 'Q2') AND
    (p.Prod_name = 'Rice' OR p.Prod_name = 'Kurta');

```

5. Pivot Operation

Purpose: To rotate the data axes and change the presentation of the result set.

Steps:

- Use SQL CASE, PIVOT, or presentation tools to rearrange the data axes.

Example (basic SQL-style pivot using CASE):

```

SELECT
    p.Prod_name,
    SUM(CASE WHEN t.month = 'January' THEN f.Total_sales ELSE 0 END) AS Jan_Sales,
    SUM(CASE WHEN t.month = 'February' THEN f.Total_sales ELSE 0 END) AS Feb_Sales FROM
    Fact_sales f
JOIN
    Product_dw p ON f.prod_id = p.prod_id
JOIN
    Time_dw t ON f.time_id = t.time_id
GROUP BY
    p.Prod_name;

```

2.4 Program and Output:

1. SLICE operation:

This query extracts only sales that occurred in the year 2024 from the sales table

SQL Worksheet

```
1 v SELECT s.BranchID, s.LocID, s.ProductID, s.TimeID, s.UnitsSold, s.DollarsSold
2 FROM Sale s
3 JOIN TIME T ON s.TimeID = T.TimeID
4 WHERE T.Yr = '2024'
5 ORDER BY s.ProductID;
```

BRANCHID	LOCID	PRODUCTID	TIMEID	UNITSSOLD	DOLLARSSOLD
3	3	1	5	10	100
3	4	1	1	10	100
3	4	1	2	10	100
3	4	1	3	10	100
3	4	1	4	10	100
3	4	1	5	10	100

This query extracts only sales of products that belong to the "Accessories" category.

SQL Worksheet

Clear

Find

Actions

```
1 v SELECT s.BranchID, s.LocID, s.ProductID, s.TimeID, s.UnitsSold, s.DollarsSold, A.Category, A.Price
2 FROM Sale s
3 JOIN Products A ON s.ProductID = A.ProductID
4 WHERE A.Category = 'Accessories' ;
```

BRANCHID	LOCID	PRODUCTID	TIMEID	UNITSSOLD	DOLLARSSOLD	CATEGORY	PRICE
1	1	2	1	10	100	Accessories	25
1	1	3	1	10	100	Accessories	75
1	1	5	1	10	100	Accessories	50
1	1	2	2	10	100	Accessories	25
1	1	3	2	10	100	Accessories	75
1	1	5	2	10	100	Accessories	50
1	1	2	3	10	100	Accessories	25
1	1	3	3	10	100	Accessories	75

2. DICE Operation:

This query retrieves sales only from the USA and limits it to Region East or West.

SQL Worksheet Clear Find Actions Save Run

```

1 SELECT s.BranchID, s.LocID, s.ProductID, s.TimeID, s.UnitsSold, s.DollarsSold, L.coun_try, B.Branchname, B.Region
2 FROM Sale s
3 JOIN Loc_dw L ON s.LocID = L.LocID
4 JOIN Branch B ON s.BranchID = B.BranchID
5 WHERE L.coun_try = 'USA'
6 AND B.Region IN ('East', 'West')
7 ORDER BY B.BranchName, L.coun_try ;

```

BRANCHID	LOCID	PRODUCTID	TIMEID	UNITSSOLD	DOLLARSSOLD	COUN_TRY	BRANCHNAME	REGION
2	1	1	1	10	100	USA	Branch B	East
2	1	2	1	10	100	USA	Branch B	East
2	1	3	1	10	100	USA	Branch B	East
2	1	4	1	10	100	USA	Branch B	East
2	1	5	1	10	100	USA	Branch B	East
2	1	1	2	10	100	USA	Branch B	East

3. Roll Up Operation:

This query calculates the total units sold (SUM(s.UnitsSold)) and the total sales amount (SUM(s.DollarsSold)) for each product (ProductName) and branch (BranchName).

```

1 SELECT
2   p.ProductName, b.BranchName, SUM(s.UnitsSold) AS TotalUnitsSold, SUM(s.DollarsSold) AS TotalDollarsSold
3 FROM
4   Sale s
5 JOIN
6   Products p ON s.ProductID = p.ProductID
7 JOIN
8   BRANCH b ON s.BranchID = b.BranchID
9 GROUP BY
10  ROLLUP (p.ProductName, b.BranchName)
11 ORDER BY p.ProductName, b.BranchName;
12

```

PRODUCTNAME	BRANCHNAME	TOTALUNITSSOLD	TOTALDOLLARSSOLD
Keyboard	Branch A	250	2500
Keyboard	Branch B	250	2500
Keyboard	Branch C	250	2500
Keyboard	Branch D	250	2500
Keyboard	Branch E	250	2500
Keyboard	-	1250	12500
Laptop	Branch A	250	2500
Laptop	Branch B	250	2500
Laptop	Branch C	250	2500

4. Pivot Operation:

This query pivots the sales data, showing total sales for each product category across different years.

```
SELECT * FROM (
  SELECT p.product_category, t.years, s.total_amount
  FROM sales s
  JOIN product p ON s.product_id = p.product_id
  JOIN time t ON s.time_id = t.time_id
)
PIVOT (
  SUM(total_amount)
  FOR years IN ('2023' AS "2023", '2024' AS "2024", '2025' AS "2025", '2026' AS "2026")
)
ORDER BY product_category
FETCH FIRST 5 ROWS ONLY
```

PRODUCT_CATEGORY	2023	2024	2025	2026
Accessories	-	-	101378	27006.6
Apparel	-	-	45677.45	11184.2
Electronics	-	-	818417.3	205853.1
Furniture	-	-	68932.7	17616.3

Download CSV

rows selected.

2.5 Conclusion:

OLAP operations like Slice, Dice, Roll-up, Drill-down, and Pivot were successfully applied to the sales data warehouse. These operations allow users to analyze data from multiple angles, making it easier to discover patterns, summarize information, and support better decision-making.

2.6 Questions:

1. What is the difference between the Slice and Dice operations in OLAP?

Aspect	Slice	Dice
Definition	Selects a single value for one dimension	Selects multiple values across multiple dimensions
Dimensions involved	One dimension	Two or more dimensions
Result	Sub-cube with reduced dimensionality	Smaller sub-cube with same dimensionality
Example	Sales data for Year = 2024	Sales for Years = 2023–2024, Regions = West/East, Product = Laptop

2. How does the Roll-up operation help in summarizing large volumes of data?

The Roll-up operation aggregates data by moving up a hierarchy or by reducing dimensions.

How it helps:

- Converts detailed data into high-level summaries
- Reduces data complexity and improves decision-making
- Enables faster analysis of large datasets

Example:

Daily sales → Monthly sales → Yearly sales

City-wise sales → State-wise sales → Country-wise sales

This allows managers to view overall trends instead of individual records.

3. Give an example of a scenario where Pivoting the data provides a clearer insight than traditional tabular view?

Scenario:

A company wants to compare **monthly sales of products across regions**.

- **Traditional table:**
Each row shows (Month, Region, Product, Sales) → difficult to compare quickly.
- **Pivoted view:**
Rows = Months
Columns = Regions
Values = Total Sales

Why pivoting is better:

- Enables **quick comparison** across categories
- Highlights trends and patterns visually
- Improves readability for business users

Example Insight:

You can instantly see which region performed best in each month, which is hard to spot in raw tabular data.